$V = IR$

$V(0) = I \times 1.923$

$R/L = 1.923$

$(s-2)^2(s+2)^2 + 9(s-2)^2$

$(s-2)(s+2)(s-2)(s+2) + 9(s-2)^2$

$(s^2-4)(s^2-4)$

$(s-2)(s+2)(s-2)(s+2)$

$v(t) = (1 - e^{-t/2})$

# EMT 3105: CIRCUIT AND NETWORK THEORY II

## CAT I

19 JUNE 2025 TIME 1:00PM

1 HOUR

ANSWER ALL QUESTIONS

a) For the transfer function given, sketch the bode diagram which shows how the phase of the system is affected by changing input frequency (5 marks)

$$TF = \frac{1}{2s + 100}$$

b) Find Laplace inverse transform of $\left\{\frac{s+2}{(s+2)^2+9}\right\}$ (5 marks)

$\frac{s+a}{(s+a)^2+\omega^2}$

c) Determine the y parameters for a two-port network if the z parameters are (5 marks)

$$Z = \begin{bmatrix} 10 & 5 \\ 5 & 9 \end{bmatrix}$$

d) Find $v(t)$ for $t > 0$, $v(0) = 5V$, $i(0) = 0$. Consider three cases in Fig 3(a):

i. Case one $R = 1.923\,\Omega$ (7 marks)

$V_0 = 5V$

ii. Case two $R = 5\Omega$

(8 marks)